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NewsBot – Executive Summary

NewsBot 2.0 is a news analysis system that reads a raw article and automatically returns useful information about it all in one place. It works like a fast reading assistant that can quickly tell what an article is about, the tone it carries, who is mentioned in it, and even translate it into another language. This project builds on my midterm, where I focused on classifying articles, and expands it into a more complete system with multiple features working together.

The system is trained on the BBC News dataset, which contains 1,490 articles across five categories: business, entertainment, politics, sport, and tech. Before any analysis happens, the text is cleaned through preprocessing by converting everything to lowercase, removing special characters and stopwords, and applying lemmatization. This step is similar to preparing ingredients before cooking—you clean and organize everything first so the final result comes out consistent and accurate.

To analyze the articles, the system converts the text into numerical form using TF-IDF, since models can only understand numbers, not raw words. It then predicts the category of the article and analyzes the sentiment to determine whether the tone is positive, negative, or neutral. Topic modeling is used to identify underlying themes in the text, while summarization generates shorter versions of long articles by selecting the most important sentences.

The system also includes named entity recognition, which works like going through a text with highlighters to pick out important names such as people, organizations, and locations. In addition, a semantic search feature allows users to find the most relevant articles based on a query. Multilingual processing is included to detect the language of a text and translate it into languages like German, French, and Spanish, which connects back to the idea that sentence structure can vary across languages.

To make the system more user-friendly, a simple chatbot interface is added so users can interact with it using natural language instead of code, almost like asking someone who has already read all the articles and can answer instantly.

Overall, NewsBot 2.0 shows how multiple NLP techniques can be combined into one system to make analyzing news faster and more efficient. While there are some limitations, such as using traditional machine learning models and a rule-based chatbot, the system is still functional and demonstrates a complete end-to-end workflow for news analysis.